

The Voxel T -Matrix

Discretising an elastic scatterer into cubes: the matched propagator,
and the accuracy of the result against exact Mie

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Abstract

A scatterer of arbitrary shape can be represented as a lattice of small cubes, each carrying an analytic single-site T -matrix, coupled through the elastodynamic propagator and solved by Foldy–Lax. This note sets out that construction, identifies the condition the propagator must satisfy for the scheme to be consistent, and measures the accuracy that results.

The condition is a pairing. A single-site T -matrix closes on a particular functional of the internal field — the value at the cube centre, or the average over the cube — and the propagator it is used with must close on the same one. The collocation closure requires the propagator averaged over the *receiver cell only*: one power of the cell form factor, not the two of a Galerkin pairing and not the none of a point evaluation.

That average is given in closed form. It is regular at every non-zero lattice separation, its far field is a scalar multiple of the point propagator per mode, and near contact it needs a quadrature order that follows the separation.

Accuracy is measured against the exact elastic Mie solution for a sphere, which is the one arbiter that is both exact and laterally varying. Two controls are needed for the comparison to measure the method rather than the test: the observation distance must reach the far field, and the staircase volume of the voxelised body must be divided out. With those in place the matched pairing is 2.9–5.8× closer to Mie than a point propagator in the Rayleigh regime. Beyond it the pairing does not help: at $ka = 0.5$ and $ka = 1.5$ the averaged propagator is slightly *worse* than the point one, and what limits the method is the staircase representation of a curved boundary.

These numbers supersede an earlier set measured against an arbiter that carried three compensating sign errors; §5 records what changed.

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1 The construction

1.1 Foldy–Lax on a lattice of cubes

Partition the scatterer into N cubes of side d on a regular lattice, with centres $\mathbf{r}_m$. Each cube is represented by a local T -matrix $\mathbf{T}_{\text{loc}}$ acting on a nine-component state — three displacements and six Voigt strains,

$$\boldsymbol{\psi} = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx})^T, \quad (1)$$

which is the smallest state closed under the coupling: the source a cube radiates depends on both the displacement and the strain it experiences.

The exciting field at cube m is the incident field plus the field radiated by every other cube,

$$\boldsymbol{\psi}_m = \boldsymbol{\psi}_m^{\text{inc}} + \sum_{n \neq m} \mathbf{P}(\mathbf{r}_m - \mathbf{r}_n) \mathbf{T}_{\text{loc}} \boldsymbol{\psi}_n, \quad \text{that is} \quad (\mathbf{I} - \mathbf{P}\mathbf{T})\boldsymbol{\psi} = \boldsymbol{\psi}^{\text{inc}}, \quad (2)$$

with $\mathbf{P}(\mathbf{s})$ the 9×9 propagator between cubes at separation $\mathbf{s}$. The composite T -matrix of the whole body follows by summing the sources, $\mathbf{T}_{\text{comp}} = \sum_m \mathbf{T}_{\text{loc}} \boldsymbol{\psi}_m$.

The self term is excluded from the sum: a cube’s action on itself is contained in $\mathbf{T}_{\text{loc}}$, which is where the singular part of the Green’s tensor and its static (Eshelby) content live.

1.2 The propagator

$\mathbf{P}$ is assembled from the elastodynamic Green’s tensor and its derivatives, in blocks

$$\mathbf{P} = \begin{pmatrix} \mathbf{G} & \mathbf{C} \\ \mathbf{H} & \mathbf{S} \end{pmatrix}, \quad (3)$$

where $\mathbf{G}$ maps force to displacement, $\mathbf{S}$ maps stress source to strain, and $\mathbf{C}, \mathbf{H}$ are the cross couplings obtained by Voigt contraction of ∂G and $\partial \partial G$. Written through the Kupradze representation, every block is built from the two scalar Helmholtz kernels g_P and g_S and their Cartesian derivative tensors, so the P and S content is separable — which §3 requires.

2 The pairing condition

2.1 Two closures

Volume-averaging the Lippmann–Schwinger equation with a uniform internal ansatz gives

$$\langle \nabla \mathbf{u} \rangle = \langle \nabla \mathbf{u}^0 \rangle + \left[\frac{1}{V} \iint \partial \partial G(\mathbf{x} - \mathbf{x}') d\mathbf{x} d\mathbf{x}' \right] \Delta c E, \quad (4)$$

a *double* average, while evaluating the same equation at the cube centre gives

$$\nabla \mathbf{u}(0) = \nabla \mathbf{u}^0(0) + \left[\int_V \partial \partial G(-\mathbf{x}') d\mathbf{x}' \right] \Delta c E, \quad (5)$$

a *single* average. Neither is wrong: one closes on the volume average of the internal field, the other on its centre value. For an ellipsoid they coincide, because Eshelby’s uniformity theorem makes the internal field uniform; for a cube they differ.

2.2 The condition

A T -matrix built on (5) must be used with a propagator that also closes on the centre value. The propagator between two cubes therefore carries *one* power of the cell form factor — the average over the receiver cell, with the source treated as a point — and not two. Writing the cell indicator’s transform,

$$\text{single average} \longleftrightarrow \text{sinc}(k_x h) \text{sinc}(k_y h) \text{sinc}(k_z h), \quad h = d/2, \quad (6)$$

against the squared factor that a Galerkin pairing produces. A point propagator carries neither and is not a consistent partner for either closure.

Per mode. The form factor must be applied to each mode separately. The P and S poles carry different k_z at the same lateral wavenumber, so a single scalar factor applied to the assembled 9×9 is incorrect.

3 The single cell average

3.1 Definition and regularity

For cubes at separation $\mathbf{s}$ on the lattice, the matched propagator is

$$\langle \mathbf{P} \rangle(\mathbf{s}) = \frac{1}{d^3} \int_{|\mathbf{u}|_\infty \leq h} \mathbf{P}(\mathbf{s} - \mathbf{u}) d\mathbf{u}, \quad h = d/2. \quad (7)$$

With a point source and a cubic receiver cell, the closest approach of the cell to the source is its near face, so

$$|\mathbf{s} - \mathbf{u}| \geq d/2 \quad \text{for every lattice separation } \mathbf{s} \neq 0, \quad (8)$$

and the integrand in (7) is regular everywhere on the domain. The double average has no such property: its integrand reaches $r = 0$ whenever the cells touch.

3.2 The far field, in closed form

The midpoint error of a cell average is

$$\langle g \rangle - g = \frac{d^2}{24} \nabla^2 g + O(d^4), \quad (9)$$

and for a Helmholtz kernel $\nabla^2 g_c = -\kappa_c^2 g_c$ exactly away from the origin. Derivatives commute with ∇^2 , so the same holds for every derivative tensor and hence for every block of (3). Therefore, away from contact,

$$\langle \mathbf{P} \rangle = \left(1 - \frac{\kappa_c^2 d^2}{24}\right) \mathbf{P} + O(d^4) \quad \text{per mode } c \in \{P, S\}, \quad (10)$$

the two modes carrying their own κ and therefore not combinable before the average is taken.

3.3 Quadrature near contact

Inside the near shell (10) is not accurate enough and (7) is integrated directly. The integrand is regular but peaked near the face of the cell closest to the source, and a Gauss rule converges slowly against a nearby peak. The difficulty is a property of the separation alone,

$$q(\mathbf{s}) = \frac{|\max(|\mathbf{s}| - h, 0)|}{d}, \quad (11)$$

componentwise in the clamp, which takes the values 0.5 for a face neighbour, $1/\sqrt{2}$ for an edge neighbour and $\sqrt{3}/2$ for a corner one. Convergence of a product Gauss rule against a 24-point reference:

neighbour	$g = 4$	$g = 8$	$g = 12$	$g = 16$
face ($q = 0.50$)	1.8×10^{-2}	2.9×10^{-5}	3.6×10^{-8}	1.4×10^{-10}
edge ($q = 0.71$)	6.5×10^{-4}	9.0×10^{-8}	1.6×10^{-10}	—
corner ($q = 0.87$)	6.1×10^{-5}	2.3×10^{-10}	—	—

Table 1. The order should follow q : 16 points per axis for face neighbours, 12 for edge, 8 for corner, and the base order elsewhere. Only six and twelve separations respectively are expensive, so the adaptive rule costs almost nothing.

Two independent evaluations of (7) — direct quadrature and the closed form (10) — agree to $O(d^4)$, halving d reducing their difference by sixteen (5.3×10^{-4} , 3.4×10^{-5} , 2.1×10^{-6} at $d = 0.2, 0.1, 0.05$), and the average reduces to the point propagator at second order as (9) requires.

3.4 Cost

$\mathbf{P}$ depends only on the separation, so on a regular lattice there are $O(n^3)$ distinct blocks rather than $O(N^2)$ pairs. Building the kernel once per distinct separation — which is what an FFT-accelerated solver does anyway — makes the averaged propagator affordable: the few near-contact separations needing high-order quadrature are evaluated once, whatever the cell count.

4 Measuring the accuracy

The exact elastic Mie solution for a homogeneous sphere is the natural arbiter: it is exact, and unlike a layered reference it is laterally varying. Two things must be controlled for the comparison to measure the discretisation rather than the test.

4.1 The observation distance

If the scattered field is compared through a far-field asymptotic on one side and an exact expression on the other, the neglected $1/(kr)$ term must be small compared with the quantity being measured. It is not a weak condition. Measured at $ka = 0.1$ with the sphere divided 4 per edge:

r/a	$k_S r$	volume-corrected error
100	10	0.5043
500	50	0.0575
5×10^3	500	0.0075
5×10^4	5000	0.0068

Table 2. The comparison is usable only from $k_S r \sim 500$; at $k_S r = 50$ it overstates the error sixfold. It is not fully converged even at the top: the last decade still moves the answer by 10%, which sets the precision of every small number in §5. The earlier version of this table read 0.0088 at both of the last two rows and so looked converged; correcting the arbiter made the errors smaller without making the harness better, and the residual $1/(kr)$ term is now a visible fraction of them.

4.2 The staircase

A cubic lattice does not fill a sphere, and its coverage is not monotonic in the number of cells per edge. For $n_{\text{sub}} = 3, 4, 6$ the voxel volume exceeds the sphere's by 26%, 4.7% and 17%, and the raw error tracks that dilution to within a factor 1.4. Two consequences follow.

First, the raw error must be divided by the volume ratio before it says anything about the propagator. Second, a refinement ladder in n_{sub} measures the geometry, not the method: the error moves non-monotonically because the dilution does. The comparison that isolates the discretisation is at *fixed* geometry — the same voxel set, the same staircase, one propagator exchanged for the other — and its stability across resolutions is what makes it meaningful.

5 Results

What changed, and why. Every number in this section is a comparison between two objects that were both wrong when it was first measured. The exact Mie solution carried a spurious $(-1)^n$ in its multipole sum, inverting the density/dipole channel; the Foldy–Lax far field carried a sign error in its force term; and the contrast-extraction route carried a third error that cancelled the first, which is how all three survived more than a thousand tests. They were found by an impedance-null test — $\delta\rho/\rho = +\varepsilon$ with $\delta M/M = -\varepsilon$ leaves $Z = \sqrt{\rho M}$ unchanged and therefore cannot backscatter — and fixed together; both solvers now agree with an independent Born calculation to 0.9997–1.0000 across $k\rho a = 0.18$ to 1.44.

Three conclusions of the earlier version do not survive the correction, and are stated here rather than quietly amended. (i) The pairing improvement at $ka = 0.5$ reversed sign: the averaged propagator was reported 1.15–1.19× better and is 0.71–0.84×, that is worse. (ii) The Rayleigh improvement was reported stable to 8.8% and has a spread of 64%. (iii) The resonance error roughly halved, from 52% to 22–28%, and refinement from $n_{\text{sub}} = 6$ to 8 now improves it where it previously made it worse. What survives is the Rayleigh result and the whole of §2–§3, which concern the propagator alone and never passed through the arbiter.

Volume-corrected far-field error against exact Mie, for a sphere with $\Delta\lambda = 2$, $\Delta\mu = 1$, $\Delta\rho = 0.1$ on a background $\alpha = 5$, $\beta = 3$, $\rho = 2.5$:

ka	n_{sub}	point propagator	matched average	ratio
0.1	3	0.0055	0.0019	2.86
	4	0.0068	0.0014	5.02
	6	0.0086	0.0015	5.76
0.5	3	0.0306	0.0365	0.84
	4	0.0141	0.0181	0.78
	6	0.0198	0.0278	0.71
1.5	6	0.2780	0.2851	0.97
	8	0.2170	0.2247	0.97

Table 3. Bold is the better of the pair. Matching the pairing helps in the Rayleigh regime and nowhere else: at $ka = 0.5$ the averaged propagator is 20–40% *worse*, consistently across resolutions, and at $ka = 1.5$ it is 3% worse. The Rayleigh improvement is also not as stable as it looked — 2.86 to 5.76, a spread of 64%, driven by the coarsest geometry.

5.1 The three regimes

Rayleigh, $ka \lesssim 0.1$. The matched average reaches 0.14–0.19%, at or below the shape-factor floor of representing a sphere by cubes, against 0.55–0.86% for the point propagator. This is

where the pairing argument of §2 is confirmed, and it is the one regime in which it is. The improvement is not uniform: $2.86\times$ at $n_{\text{sub}} = 3$ against $5.76\times$ at $n_{\text{sub}} = 6$. The coarsest case is also the most diluted (26%), so the spread is plausibly geometric rather than a property of the propagator — but that is an explanation offered, not one tested, and the earlier claim of stability across resolutions is withdrawn.

Transition, $ka \sim 0.5$. **The pairing stops helping and starts hurting:** $0.71\text{--}0.84\times$, consistently across all three resolutions. The effect is systematic, not scatter, and it is larger than the harness uncertainty of the previous subsection. Two readings are available and this note does not choose between them. Either the collocation closure that §2 matches is itself losing validity as ka_{sub} grows, so that matching it matches the wrong thing; or the cell average is correct and is being scored against a staircase error that happens to cancel part of the point propagator's own. Distinguishing them needs a body whose boundary the lattice resolves exactly — a cube, not a sphere — which is a different measurement from this one.

Resonance, $ka \gtrsim 1$. The error is 22–28%, and the propagator is not what is wrong: the two propagators differ by 3% where the error is 25%. The sub-cell T -matrices remain inside their Rayleigh validity ($ka_{\text{sub}} = 0.25$ at $n_{\text{sub}} = 6$), and refining from $n_{\text{sub}} = 6$ to 8 now improves the error from 27.8% to 21.7% as the dilution falls from 17% to 4.3% — so what the ladder is tracking here is still the staircase, not the method. What fails is the representation of a curved boundary at a wavelength comparable with the radius. That is a limit of the discretisation itself, not of the propagator within it.

6 Summary

1. Partition the body into cubes; give each the analytic single-site T -matrix for its contrast.
2. Couple them with the propagator averaged over the *receiver* cell, (7), applied per mode — matching the collocation closure of the single-site T -matrix.
3. Use the closed form (10) beyond a near shell and direct quadrature inside it, with the order chosen from (11).
4. Solve (2); build the kernel once per distinct separation.
5. When comparing against an exact reference, reach the far field and divide out the staircase volume; compare at fixed geometry.

The method is accurate to a fraction of a percent in the Rayleigh regime, degrades to one to four percent through the transition, and is geometry limited beyond $ka \sim 1$ at some twenty-five percent, where a representation that follows the boundary rather than staircasing it is required instead.

The pairing of step 2 is what the Rayleigh figure rests on, and it is established only there. At $ka = 0.5$ and above the averaged propagator is measurably worse than the point one, and until that is understood step 2 should be read as a Rayleigh-regime recommendation rather than a general one.